Loop Engineering with LangChain DeepAgents

Overview of Loop Engineering





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Prompt, Context, Harness, and Loop Engineering

Prompt Engineering

Shapes what you ask the model — the instructions that guide its behavior and output.

Context Engineering

Shapes what the model knows at each step — selecting and organizing the right information, memory, and tool results.

Harness Engineering

Shapes the system around the model: tools, middleware, permissions, workflows, and other infrastructure that determines how the agent operates.

Loop Engineering

Shapes how the system repeatedly acts, evaluates, and improves: designing feedback loops that enable agents to complete work, verify results, and improve over time.

Why loops matter

Why loops?

- One model call is rarely enough for complex work
- Agents need to act → observe → decide → act again
- Repeated execution creates opportunities for feedback and improvement

Continuous improvement

- Every run produces rich execution outcomes and traces
- Outcomes directly transition into actionable feedback
- Feedback loops continuously refine and improve future behavior

Four levels of loops

1) Agent loop

Automate fundamental workflows

2) Verification loop

Improve end-to-end reliability

3) Event-driven loop

Automate work in real-world systems

4) Hill-climbing loop

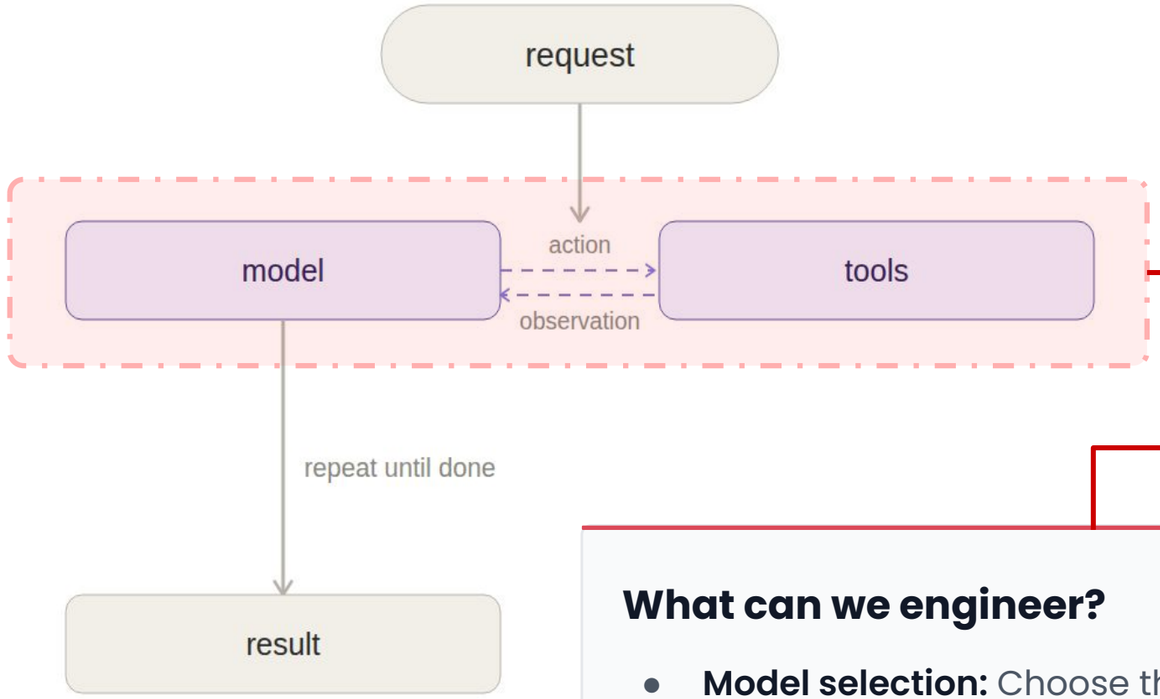
Optimize and improve the agent itself

Loop 1: The Agent Loop

A basic agent loop: A model repeatedly interacts with tools until it can complete a request.

Execution steps

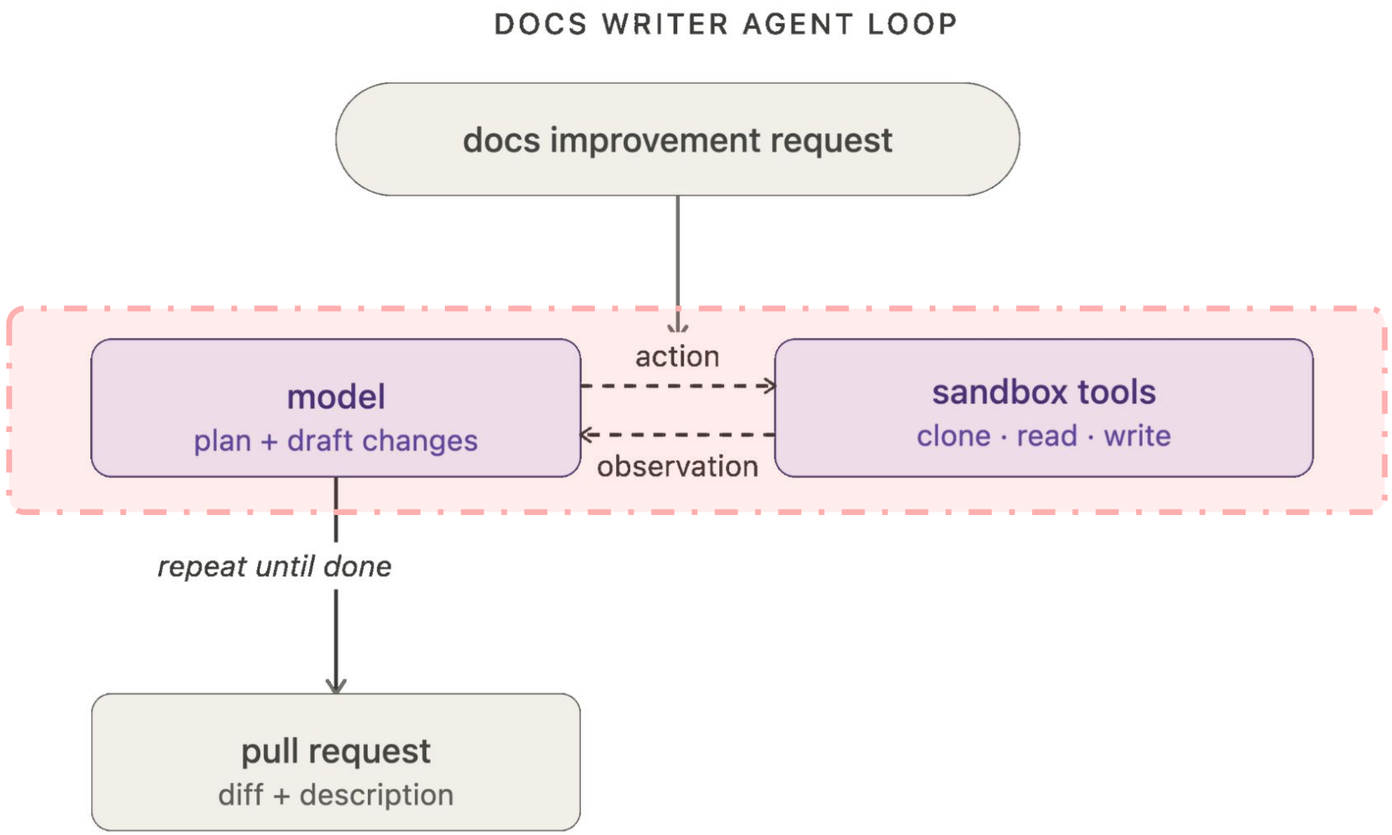
- Task + context
- Model reasons about next action
- Model calls a tool
- Tool returns an observation
- Model chooses next action
- Repeat until done



What can we engineer?

- **Model selection:** Choose the right capability
- **Tool selection:** Provide relevant capabilities
- **Tool descriptions:** Optimize definitions

Agent Loop - Example



Level 2: Verification loop

Verification Flow

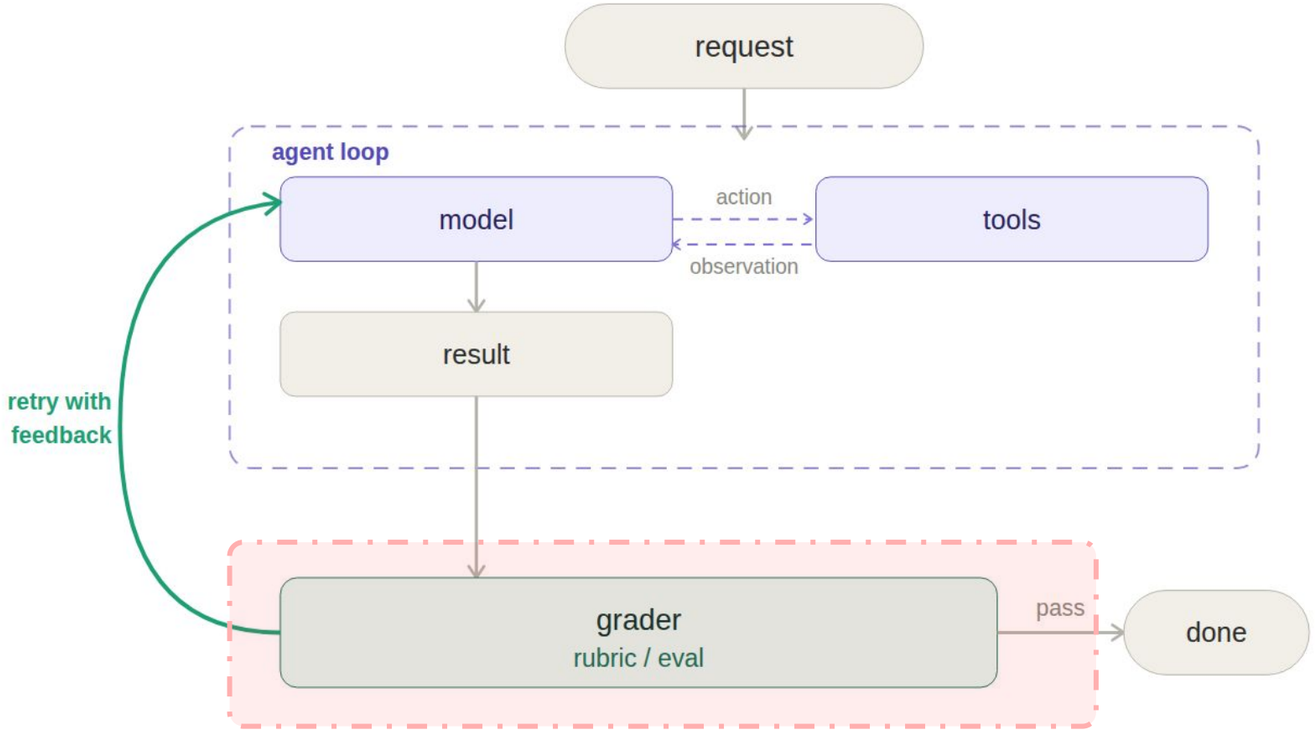
- **Request:** Initial system input.
- **Agent Attempt:** Executes the task.
- **Grader Evaluation:** Result sent to grader.
- **Feedback Loop:** Fail feedback retry.
- **Done:** Pass finished.

Why verification helps

- Explicit quality criteria
  - Higher reliability
  - Fewer incomplete results
- **Can make cheaper models viable for harder tasks**

Grader checks the agent's output against a rubric and, if it fails, sends the result back with feedback.

Type: Graders can either be deterministic or agentic (LLM as a judge is a classic example, here).



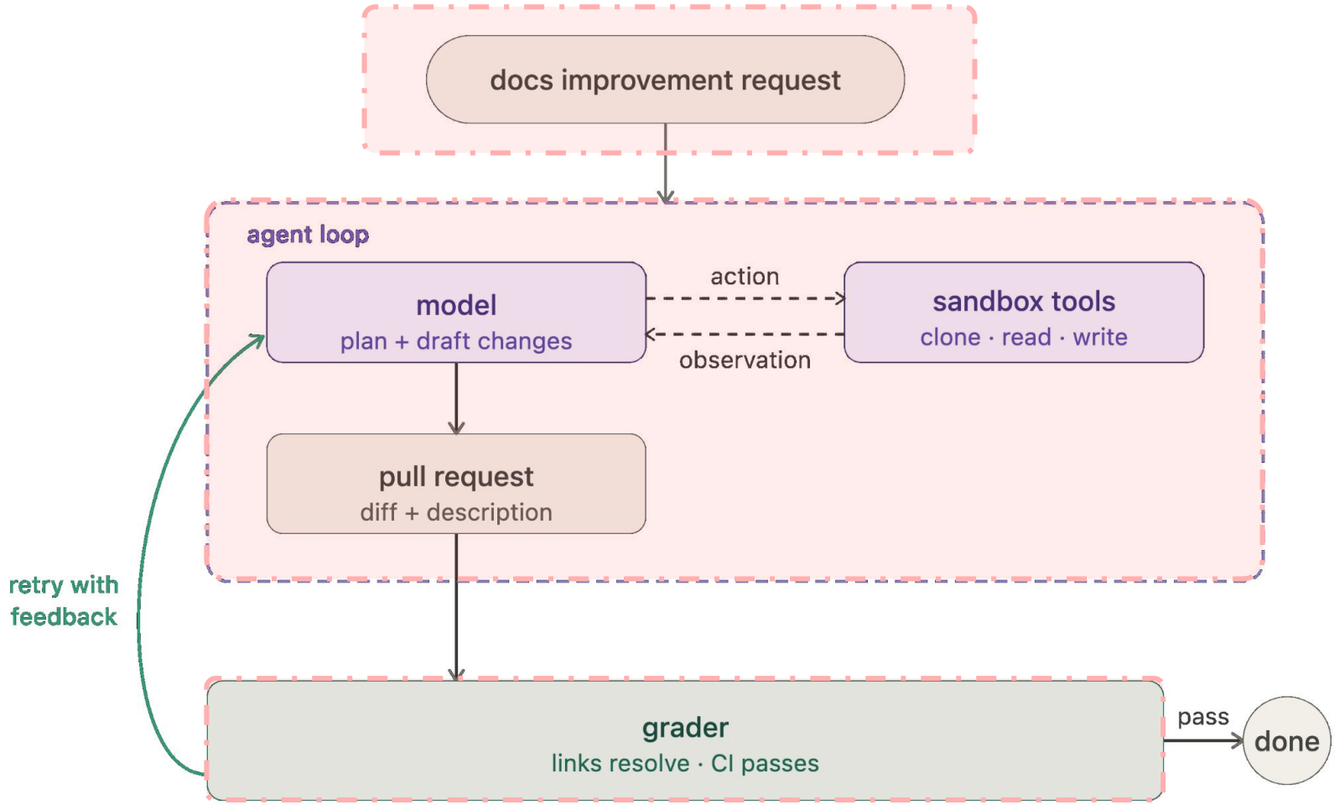
Level 2: Verification loop – Example

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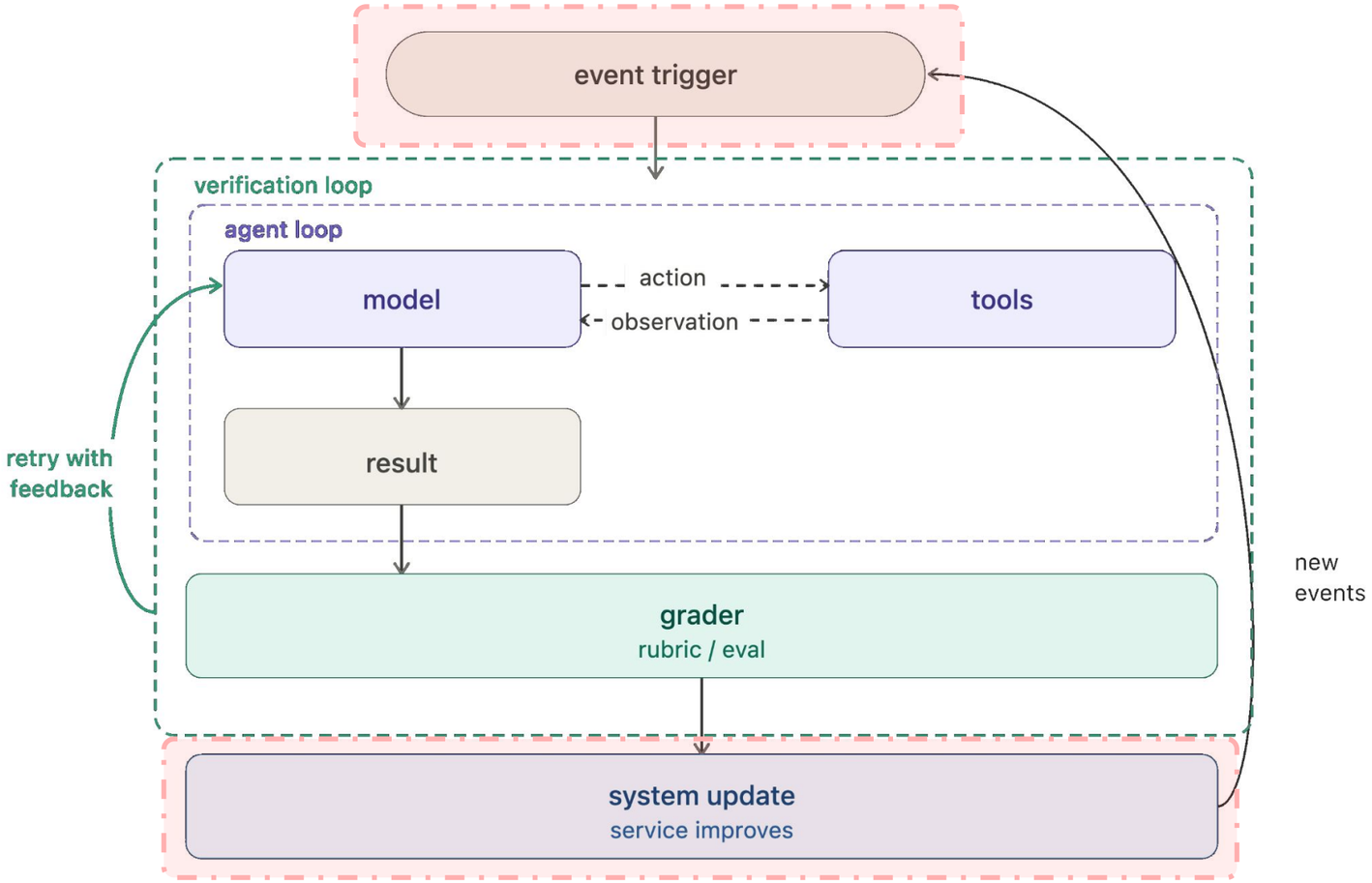
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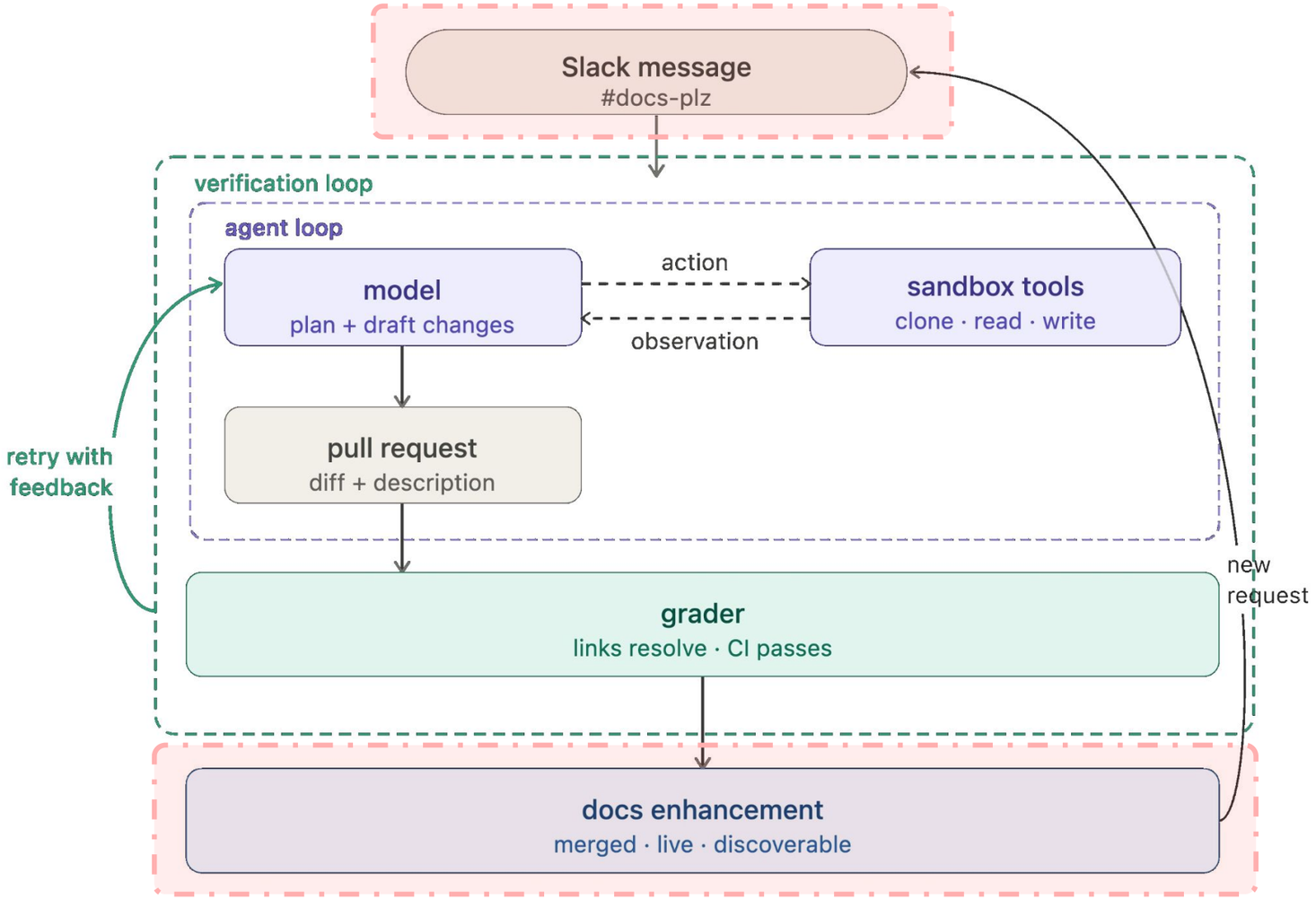
Level 3: Event driven loop

The event-driven loop connects your agent to your ecosystem.

- An event fires
- A new document lands
- A schedule triggers
- A webhook arrives



Level 3: Event driven loop – Example

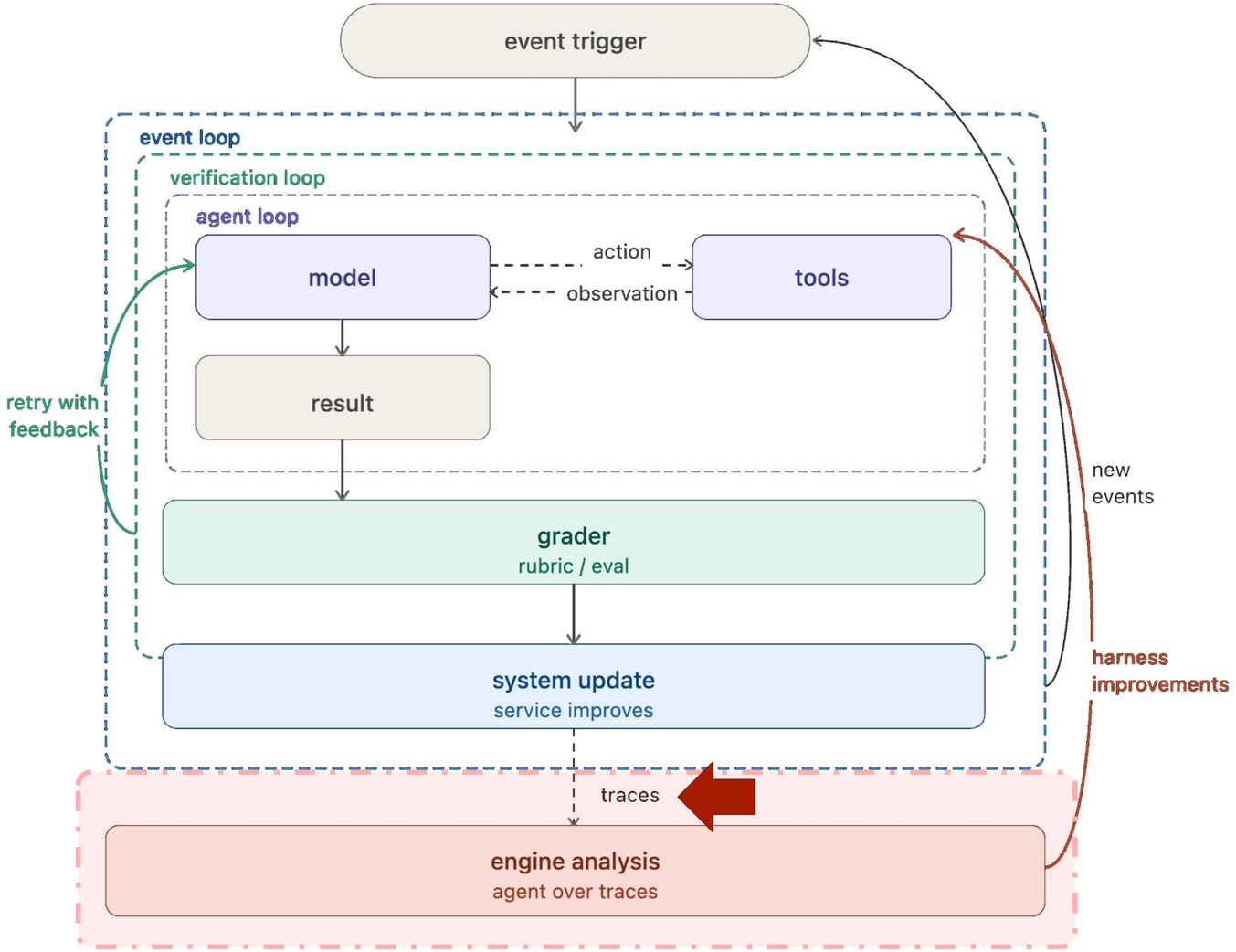


Level 4: Learning / hill climbing loop

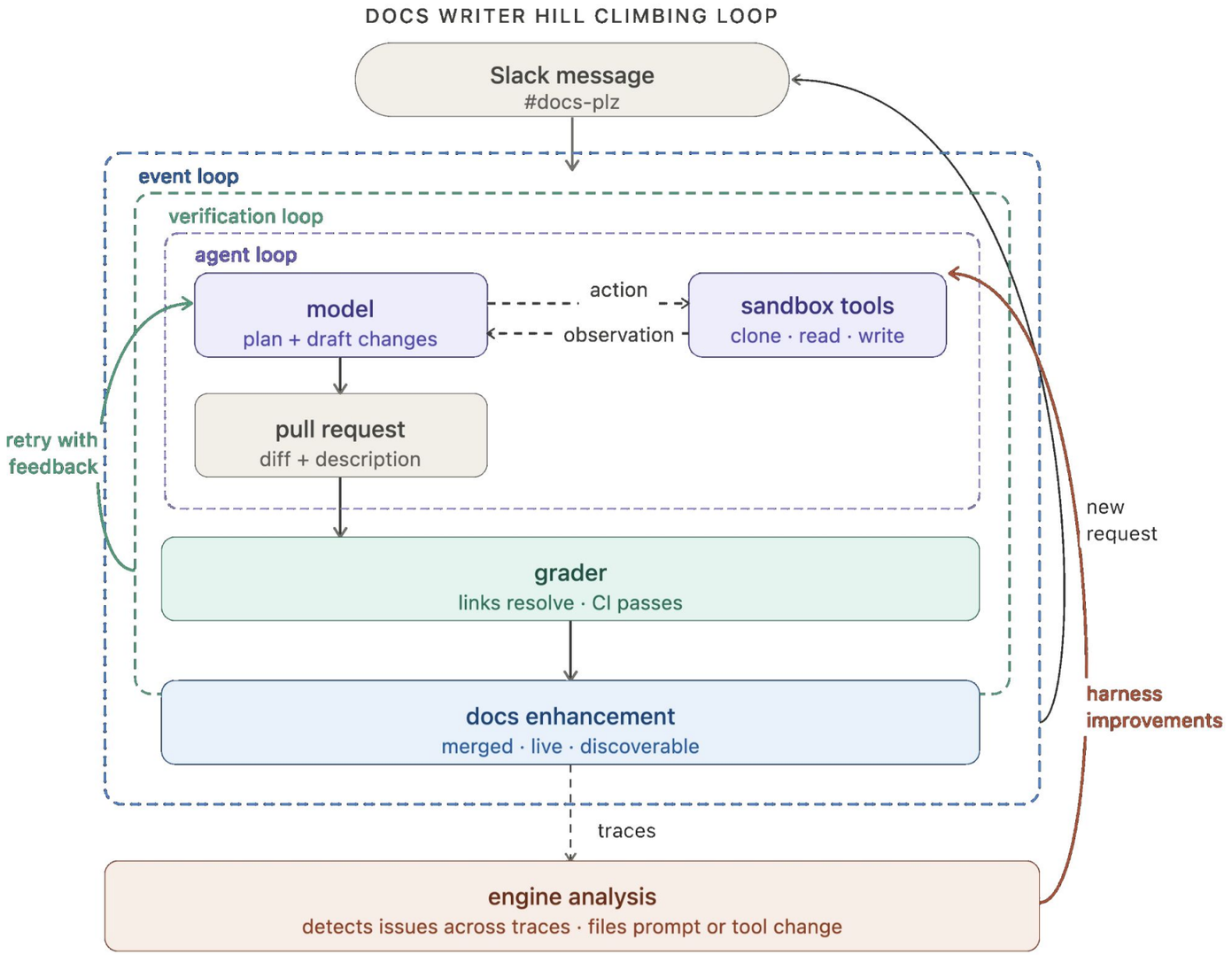
Trace: a record of what the model did, the tools it called, grader feedback, etc.

The hill climbing loop:

- improve base procedural things
- more memory information



Level 4: Learning / Hill Climbing Loop- Example



Human oversight and expertise

1. AGENT LOOP

Require human input before executing sensitive tools or final actions.

2. VERIFICATION LOOP

A human acts as the grader to audit and score sensitive automated workflows.

3. APPLICATION LOOP

Human-in-the-loop approves raw outputs before final delivery to users.

4. HILL CLIMBING LOOP (System-level Optimization)

Harness and evaluation improvements must flow through systematic human review before final system deployment.

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Putting it all together

Loop	What is looping?	Core idea	Result
1. Agent loop	Model ↔ tools	Act, observe, act again	Task completed
2. Verification loop	Agent run ↔ evaluator	Evaluate the result and retry with feedback	Better-quality task result
3. Event-driven loop	Event ↔ agent execution	Run agents whenever something happens	Ongoing automation
4. Hill-climbing loop	Production traces ↔ harness	Learn from many runs and improve the system	Better agent itself